ANSWER KEY & MARKING SCHEME SET A

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** vegetable peels	Broken down by microbes
2	**(b)** producers	First trophic level
3	**(c)** 10% of the previous level	Lindeman
4	**(a)** Correct	Stratosphere
5	**(a)** Student A is correct	Refrigerants
6	**(d)** biological magnification	Non-biodegradable
7	**(a)** crop field	Man-made
8	**(d)** decomposers	Bacteria and fungi
9	**(d)** tertiary/secondary consumer (carnivore)	Eats herbivore
10	**(a)** UV radiation	$O_2 \longrightarrow O + O$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A functional unit consisting of all living organisms (biotic) of an area interacting with the non-living (abiotic) components.	2
12	They break down dead organisms and wastes, returning nutrients to the soil (recycling).	2
13	Plastic, DDT, glass, aluminium cans (any two)	1+1
14	False — only 3–4 levels exist because energy decreases greatly at each level.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Only about 10% of energy passes to the next trophic level; e.g. grass (1000 J) → deer (100 J) → lion (10 J)	1+2
16	$O_2 \xrightarrow{UV} O + O$; $O + O_2 \longrightarrow O_3$; UV radiation can cause skin cancer, cataract and harm plants; CFCs	1+1+1
17	Accumulation of non-biodegradable chemicals at increasing levels in each trophic level; humans occupy the top of food chains	2+1
	OR: Segregation of biodegradable and non-biodegradable waste; composting; reduce, reuse and recycle; avoid plastic bags	3
18	Energy available decreases at each level (90% lost as heat); after 3–4 levels little energy remains	3
	OR: Biodegradable: broken down by microbes (paper, peels); non-biodegradable: not broken down, persist (plastic, DDT); latter cause pollution	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) network of interconnected food chains (ii) Sun → producers → herbivores → carnivores; about 10% transferred at each level; rest lost as heat (iii) energy captured by autotrophs does not revert to the Sun; energy passed to herbivores does not come back	1+2+2
	OR: (i) steps of a food chain where transfer of energy occurs (ii) grass → grasshopper → frog → snake (iii) biotic: producers, consumers, decomposers; abiotic: temperature, rainfall, soil, minerals	1+2+2
20	(i) biological magnification (ii) fish-eating birds (iii) highest trophic level; the chemical is non-biodegradable and accumulates (iv) use bio-pesticides/limit chemical pesticides	1+1+2+1
	OR: (i) layer of O_3 in the upper atmosphere (ii) CFCs used as refrigerants and in fire extinguishers (iii) UNEP agreement (1987) to freeze CFC production at 1986 levels (iv) skin cancer	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** grass, goat and human	Producer → consumer
2	**(a)** grass, flowers and leather	No plastic/glass
3	**(b)** producers	Autotrophs
4	**(c)** Incorrect — wastes would pile up and nutrients would not be recycled	Recycling
5	**(b)** Student A is correct	Solar energy
6	**(a)** 1%	Converted to food
7	**(c)** three oxygen atoms	O_3
8	**(d)** bacteria	Also fungi
9	**(b)** 100 J	10%
10	**(b)** artificial ecosystem	Man-made

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The various steps in a food chain at which transfer of food/energy takes place.	2
12	Decomposers are absent or few, so wastes accumulate.	2
13	Temperature, rainfall, soil, water, minerals (any two)	1+1
14	Not fully — making kulhads on a large scale would cause loss of fertile top-soil.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Food chain is disturbed: organisms of the next level starve; the previous level grows unchecked; ecological imbalance	3
16	Removing producers kills all other levels; removing top carnivores increases herbivores which overgraze; impact depends on position	3
17	Shields Earth from harmful UV radiation; $O_2 \longrightarrow O + O$ by UV, then $O + O_2 \longrightarrow O_3$	1+2
	OR: Use of disposable items increases non-biodegradable waste; carry cloth bags; segregate waste; compost kitchen waste	1+2
18	Less waste, saves resources and energy, more hygienic if cleaned properly (kulhads vs plastic cups), cheaper in the long run	3
	OR: They make food from sunlight (autotrophs); all other organisms depend on them directly or indirectly	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) sequence of organisms eating one another (ii) phytoplankton → zooplankton → small fish → large fish; grass → deer → lion (iii) dead matter accumulates, nutrients are not recycled	1+2+2
	OR: (i) increase of harmful chemicals at higher trophic levels (ii) sprayed on crops, washed into soil/water, absorbed by plants (iii) humans are at the top of many food chains (iv) DDT	1+2+1+1
20	(i) biodegradable: food waste; non-biodegradable: plastic, glass (ii) composting/biogas for food waste; recycling plastic and glass (iii) releases toxic gases (iv) reduce, reuse, recycle	1+2+1+1
	OR: (i) only ~10% of energy passes to the next level (ii) producer 10 000 → primary 1000 → secondary 100 → tertiary 10 J (iii) less energy available	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.